

1.



SIMATS
ENGINEERING

SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS 16 / 20 completed

Online Reservation ▾

ONLINE RESERVATION SYSTEM

Online Reservation System: Create a class for an online reservation system (e.g., for flights, concerts) with methods to book, cancel, and display reservations.

PROGRAM EDITOR C++ Run Save

INPUT

Alice
AI101
Bob
BA202
Alice

2.



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QUESTIONS 16 / 20 completed

Social Media Profile ▾

SOCIAL MEDIA PROFILE

Social Media Profile:
Design a class for a social media profile, including attributes for user information and methods to add posts, manage friends, and display the user's profile.

PROGRAM EDITOR

C++ ▾

Run Save

```
1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class UserProfile {
8 private:
9     string username;
10    string email;
11    vector<string> friends;
12    vector<string> posts;
13
14 public:
15     UserProfile(string u, string e) : username(u), email(e) {}
16
17     void addFriend(string friendName) {
18         friends.push_back(friendName);
19         cout << friendName << " added as a friend." << endl;
20     }
21 }
```

INPUT

john_doe
john@example.com
Alice
Hello world! This is my first post.

3.

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QUESTIONS 16 / 20 completed

Hospital Managemen ▾

HOSPITAL MANAGEMENT SYSTEM

Hospital Management System: Construct a class system for managing a hospital, including classes for patients, doctors, and appointments. Implement functionalities to schedule appointments and manage patient records.

PROGRAM EDITOR C++ Run Save

INPUT

John Doe
101 30
Dr. Smith
201
Cardiology
2026-08-15

4.



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QUESTIONS 20 / 20 completed

Inventory Tracking ▾

INVENTORY TRACKING SYSTEM

Inventory Tracking System: Develop a class for inventory tracking in a warehouse or store, with functions for tracking items, managing stock levels, and displaying item details.

PROGRAM EDITOR

C++ ▾ Run Save

```
1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6 class InventoryItem {
7 private:
8     string itemID;
9     string itemName;
10    int stockLevel;
11
12 public:
13     InventoryItem(string id, string name, int initialStock) {
14         itemID = id;
15         itemName = name;
16         stockLevel = initialStock;
17     }
18
19     void updateStock(int amount) {
20         stockLevel += amount;
21         cout << "Stock updated." << endl;
22     }
23 }
```

OUTPUT

```
Item Details:
Item ID: INV1001
Item Name: Wireless Mouse
Current Stock: 50

Stock updated.
Item Details:
Item ID: INV1001
Item Name: Wireless Mouse
Current Stock: 70
```

INPUT

INV1001
Wireless Mouse
50
20

5.



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QUESTIONS 20 / 20 completed

Restaurant Orderin

RESTAURANT ORDERING SYSTEM

Restaurant Ordering System: Create a class system to manage orders in a restaurant, including classes for menu items, customer orders, and bill generation.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <string>
3 #include <vector>
4
5 using namespace std;
6
7 class MenuItem {
8 public:
9     string name;
10    double price;
11
12    MenuItem(string n, double p) : name(n), price(p) {}
13 };
14
15 class Order {
16 private:
17     vector<MenuItem> items;
18
19 public:
20     void addItem(string name, double price) {
21         items.push_back(MenuItem(name, price));
22     }
23 }
```

OUTPUT

```
Item Added: Burger, Price: 8.5
Item Added: Pizza, Price: 12

--- Restaurant Bill ---
Item: Burger | Price: 8.5
Item: Pizza | Price: 12
Total Bill: 20.5
```

INPUT

```
Burger
8.5
Pizza
12.0
```